

6.8 Integer Subtraction

Lesson Introduction

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 Benchmark	MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.		 Learning Target	I can subtract integers with values greater than 100.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.		MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- How do you rewrite a subtraction-of-integers problem as addition of the additive inverses? - How do you subtract integers with values greater than 100?			
 Lesson Narrative	In this lesson, students develop procedural fluency with subtracting integers greater than 100. Students explore how subtraction can be rewritten as adding the additive inverse. Students write and solve subtraction of integers problems based on real-world elevation scenarios.			
 Component Summary	6.8.1 Warm-Up (~5 min.)	Reflect on mastery of skills related to subtracting integers (<i>SE p. 270</i>)	6.8.3 Try It (10-15 min.)	Practice subtracting integers in the context of elevation (<i>SE p. 271</i>)
	6.8.2 Guided Practice (15 min.)	Explore the equivalence of subtracting and adding the additive inverse (<i>SE p. 270</i>)	6.8.4 Your Turn! (10 min.)	Additional practice subtracting integers (<i>SE p. 272</i>)
	6.8.5 Wrap-Up (~5 min.)	Demonstrate understanding of lesson learning target (<i>SE p. 273</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Existence of additive inverse <u>Additional Terms:</u> N/A		Chart paper & markers	
		<u>Additional Preparation</u> Prepare an anchor chart with visual models, instructions, and examples.		

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to represent a negative change as a subtraction expression. - Students may struggle justifying why expressions are equivalent. - Students may struggle representing subtraction equations as addition equations using the additive inverse.	- Consider providing sentence starters for the written responses for 6.8.2 Guided Practice. - Show students how to create open number lines, because of the size of the integers, to model the problems in 6.8.2 Guided Practice.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 6.8.2 Guided Practice provides opportunities for students to initially try the problems independently and then discuss their work with a partner. The discussions will help students gain confidence and understanding.	MA.K12.MTR.4.1: Error Analysis 6.8.2 Guided Practice provides an opportunity for students to analyze samples of student work. Students discuss the errors and how to correct them. These discussions help clear up common misconceptions and deepen student understanding of subtracting integers.
 Differentiate Instruction	6.8.3 Try It and 6.8.4 Your Turn! both provide practice for what serves as the culmination of three lessons on subtracting integers before students transition to multiplication. This practice is an opportunity for students to engage in a variety of tools and resources that provide differentiated support.	
	Consider utilizing On-Ramp to 6 th Grade Learning Tool to provide scaffolding for students in need of additional support subtracting integers. - Addition and Subtraction Fluency - Number Lines: Whole-Number Sums and Differences	
Lesson Notes:		

6.8.1 Warm-Up: Self-Reflection

Component Overview: Students reflect on their mastery of skills related to subtracting integers. Consider having students look back at the Wrap-Up formative assessments from previous lessons to see examples of each statement. Seeing the examples will help students make more valid and reliable ratings.



6.8.1 Warm-Up: Self-Reflection

- For each statement, rate yourself on a scale of 1-5. A rating of 5 means “absolutely yes” and 1 means “absolutely not.”

Statement	Rating
I can subtract integers using integer chips.	
I understand how to show subtraction of integers on the number line.	
I can subtract integers without using integer chips or a number line.	
I can explain to someone else how to subtract integers.	

Answers will vary.



Consider utilizing these ratings or reflections to inform your differentiation in 6.8.3 Try It or 6.8.4 Your Turn!

6.8.2 Guided Practice: Subtracting Integers

Component Overview: With a partner, students determine if a subtraction and addition expression are equivalent. Students write an addition and a subtraction expression to represent changes in scores. Students analyze two sample students' answers, identifying and correcting errors. Finally, students consolidate their understanding that a subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to addition and changing the second value to its opposite sign.



6.8.2 Guided Practice: Subtracting Integers

1. Two expressions are shown.

Expression A	Expression B
$-191 + 77$	$-191 - (-77)$

With your partner, determine if the expressions are equivalent. Summarize your determination below.

Answers will vary. The expressions are equivalent.

If thinking with integer chips in mind:

Expression A: In this expression, there is -191 and $+77$. Eliminate the 77 zero pairs formed when combining these values, and there will be -114 remaining.

Expression B: In this expression, there is -191 , and -77 is being removed. There will be -114 remaining.

OR

The expressions are equivalent.

If picturing a number line:

Expression A: If using a number line, I would start at 0 and move 191 units to the left to represent -191 . From there, I would move 77 units to the right. The final location will be at 114 on the negative side of the number line.

Expression B: If using a number line, I would start at 0 and move 191 units to the left to represent -191 . I would turn around from -191 because of the double negative and move 77 units to the right. The final location will be at 114 on the negative side of the number line.

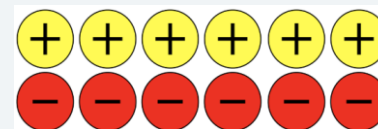


MA.K12.MTR.3.1: Fluency

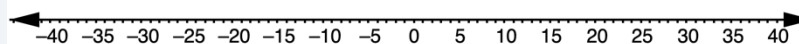
Monitor for students who share different approaches (as seen in the answer key) and elicit those responses whole group to highlight the flexibility of fluency in problem-solving.



The first sample response in the answer key would be for students thinking about integer chips to explain their answer.



The second sample response in the answer key would be for students thinking about a number line to explain their answer.



6.8.2 Guided Practice: Subtracting Integers (continued)

2. Annalise is on a trivia game show. Her score after the first round is -500 points. She misses the first question in the second round, losing an additional 125 points.

Kiersten said she could represent Annalise's score using subtraction, while Dylan said it could only be represented with addition.

- a. Complete the table with the expression each student was thinking of.

Kiersten's Subtraction Expression	Dylan's Addition Expression
$-500 - 125$	$-500 + (-125)$

- b. Kiersten said Annalise's score was -375 after missing the first question in the second round, but Dylan said it was -625 . Explain which student is correct.

*Answers will vary. In this scenario, Annalise **lost** 500 points in the first round. Then she **lost** 125 points in the second round.*

Dylan is correct. Because Annalise lost points in two consecutive rounds, she should have more negative points after the second round. Dylan added two negative integers to indicate the loss of points. The result is a negative number less than the values of the points lost in either of the two rounds.

Kiersten's expression, $-500 - 125$ indicates that 125 positive integers are to be subtracted from 500 negative integers.

Her answer, however, indicates that she subtracted 125 negative integers from the 500 negative integers. Her expression is correct, but her answer is not.



Consider having students who finish early explain why it is possible to represent one scenario as both a subtraction and addition expression and why they will be equivalent. Students should mention concepts of multiple representations (MA.K12.MTR.2.1) or provide examples.



If students struggle to start writing a response, provide them with sentence starters or a scaffolded fill-in-the-blank paragraph. Alternatively, prompt students to participate in the Stronger and Clearer language routine to structure responses.

6.8.2 Guided Practice: Subtracting Integers (continued)

3. Rewrite each subtraction equation as an equivalent addition equation.

Subtraction Equation	$513 - 382 = 131$	$957 - 136 = 1,093$
Addition Equation	$513 + (-382) = 131$	$957 + 136 = 1,093$

Addition and subtraction are **inverse operations**, which means they are operations that have the opposite effect of each other.

Understanding this relationship and the existence of additive inverses can be helpful when subtracting integers.



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

4. Complete the statements below using this information.

A subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to **addition** and changing the second value to its **additive inverse**.



Activate prior knowledge by referencing the integer chips table of representations.



- What happens when you add additive inverses?
- How do you know they will always equal zero?
- How can additive inverses or zero pairs help us when subtracting integers?

6.8.3 Try It: Subtracting Integers

Component Overview: Students write and solve expressions to represent the changes in elevation. Students also explain the meaning of negative elevations. Finally, students practice subtracting integers.



6.8.3 Try It: Subtracting Integers

- The table shows the beginning and ending elevations, in feet (ft), of six different hiking trails. The change in elevation along the trail can be found by subtracting the initial elevation from the final elevation.

$(\text{final elevation}) - (\text{initial elevation}) = (\text{change in elevation})$

- Write an expression that represents the change in elevation along each hiking trail. Then, write the value that represents the change in elevation. The first row has been completed as an example.

Initial Elevation (ft)	Final Elevation (ft)	Difference Between Final and Beginning	Change in Elevation (ft)
400	900	$900 - 400$	500
115	68	$68 - 115$	-47
294	-121	$-121 - 294$	-415
-200	610	$610 - (-200)$	810
-163	-57	$-57 - (-163)$	106
-17	-87	$-87 - (-17)$	-70

- Explain what a negative change in elevation means in this context.

Answers will vary. A negative change in elevation means that the hiker's final elevation is less than the initial elevation. It also means that a hiker moved down from, or below his or her initial elevation.

- Find each difference.

- $-664 - 231$
-895

- $109 - (-115)$
224



To activate prior knowledge, consider reviewing how to represent positive and negative elevations. Prompt students to explain or represent concepts like sea level or mountains using vertical number lines.



How did you find the change in elevation?
If a hiker is moving down the hiking trail, will there be a positive change in elevation or a negative change in elevation? Why?



Is it possible for a hiker to be at a negative elevation?
Is that a reasonable (MA.K12.MTR.6.1) solution? Why or why not?

6.8.4 Your Turn!

Component Overview: Students have additional practice subtracting three-digit integers.



6.8.4 Your Turn!

1. Evaluate each expression.

a. $740 - (-171)$

911

b. $569 - 143$

426

c. $-738 - (-212)$

-526

d. $333 - (-204)$

537

e. $-877 - 814$

-1691

f. $-691 - 293$

-984

g. $157 - (-120)$

277

h. $341 - (-504)$

845



Consider methods for differentiated support during this activity based on your students' performances and your informal observations during Lessons 6-8.

Station Rotations

- **Station 1:** Teacher-led small group (for additional support)

- **Station 2:** On-Ramp* video and practice (for additional support)

- **Station 3:** Your Turn! independent practice (for on-level benchmark practice)

- **Station 4:** Desmos* Activity (for application and extension)

Students can either opt-in to each station as they need it, opting in to On-Ramp if they want more support with subtraction fluency from a Study Expert or trying some practice problems first before checking in to a teacher-led small group. Alternatively, students can be placed in three groups based on their responses from the 6.8.1 Warm-Up in this lesson or the Wrap-Up from the previous lesson.

*On-Ramp and Desmos are available in multiple languages for added scaffolding and can be found in the Teacher Area.

6.8.5 Wrap-Up: Ticket Out the Door

Component Overview: Students evaluate three-digit integer subtraction expressions. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.



6.8.5 Wrap-Up: Ticket Out the Door

1. Evaluate each expression.

a. $447 - (-702)$

1149

b. $-834 - (-610)$

-224



Provide blank number-line models for students to utilize if necessary.

c. $-309 - 525$

-834



Challenge students to choose one question and create a math story similar to question 1b from 6.8.3 Try It.